

Toward Supporting Narrative-Driven Data Exploration: Barriers and Design Opportunities



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prototype system



poster summary

Exploration rarely goes as planned. A single chart leads to a new question, an unexpected pattern shifts the focus. And soon, a simple check becomes a deeper search for meaning.

To understand what holds analysts back from this kind of narrative-driven inquiry, we surveyed 48 data practitioners about how their questions evolve during open-ended analysis.

What's preventing deeper engagement?

Observed User Barriers



Narratives Drift Toward Bias
Analysts risk reinforcing early assumptions and overlooking contradictory evidence.



No Built-In Space to Externalize Thinking
Users try to track evolving thoughts, but there's nowhere to keep them visible.



Reasoning paths are easily lost
Many "lost the thread" after interactions, existing tools does not capture the semantic logic that links one insight to the next.



Fragmented Views Block Cross-View Reasoning
When trying to connect or synthesize, everything is scattered, insights remain siloed.

What kinds of support would help?

Design Implications



Scaffold evolving hypotheses as branchable, elements within exploration



Enable multi-view synthesis across disconnected data perspectives



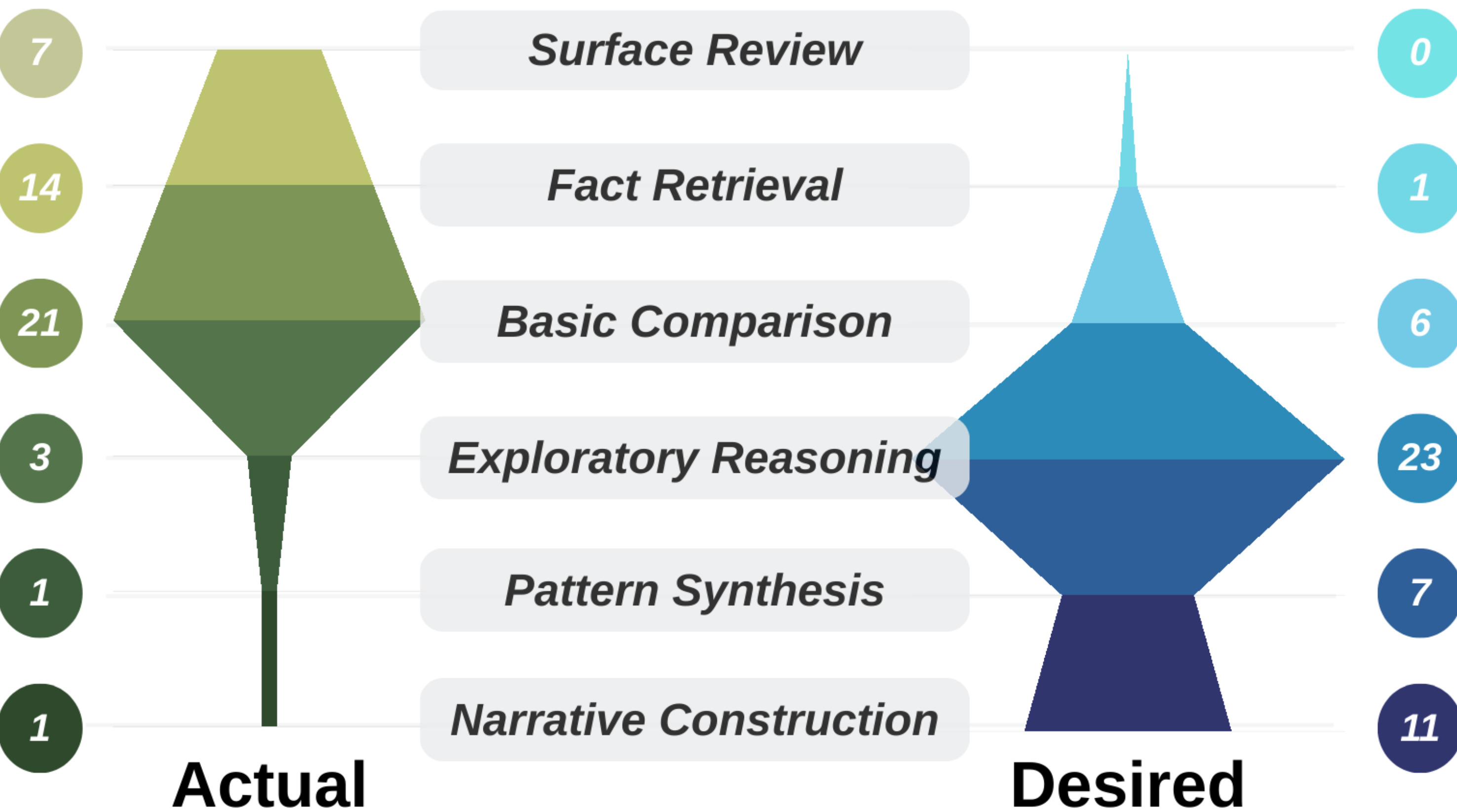
Preserve reasoning paths using insight-level provenance and semantic cues



Promote reflective exploration and alternative perspectives

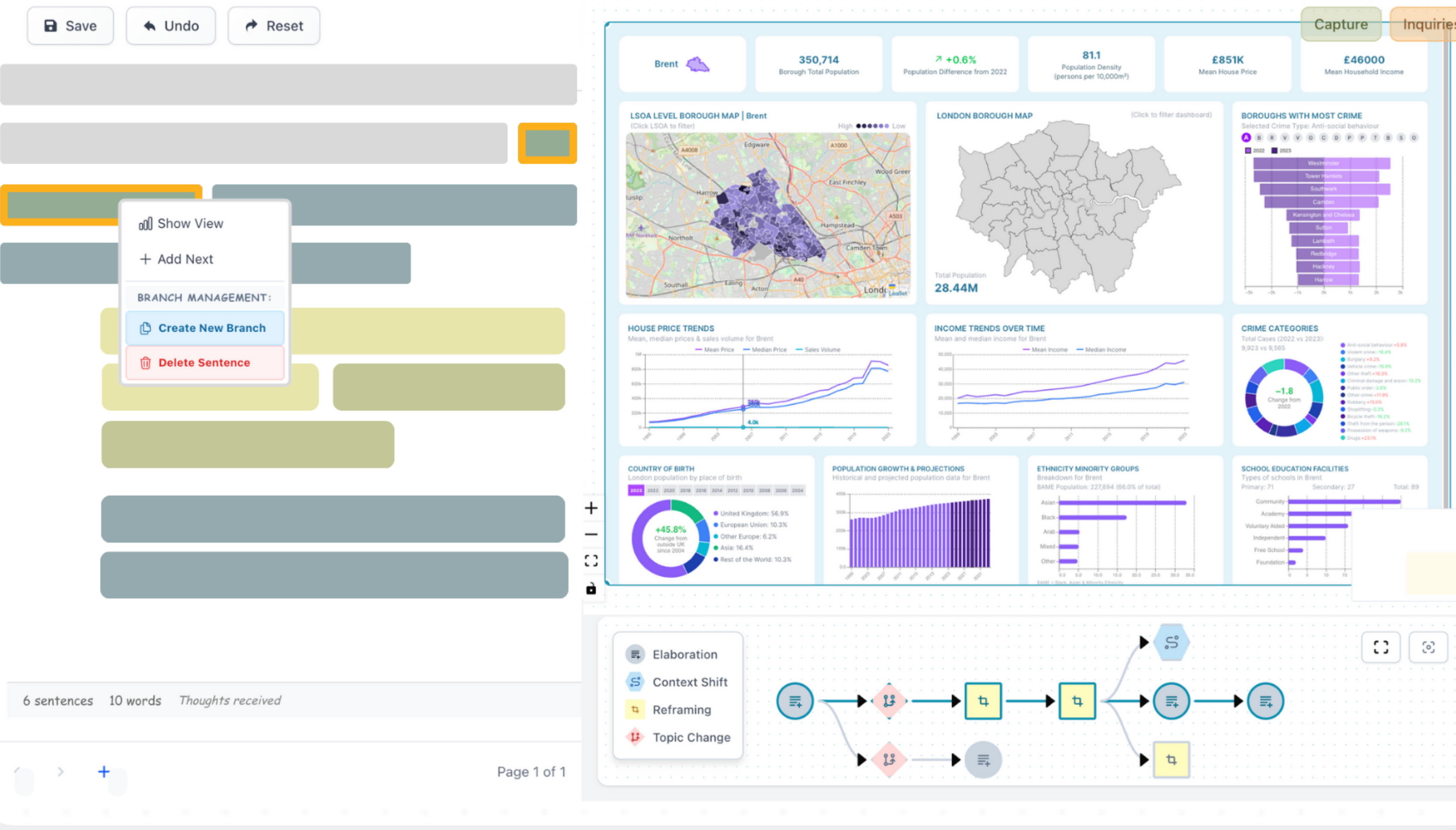
Engagement Level Overview

*From our semi-structured survey, we found **six** levels of engagement, from surface-level monitoring to constructing and externalizing coherent narratives.*



How might a new workflow support these?

Visual Workflow: Narrative Scaffolding



[1] A. Batch and N. Elmquist. The interactive visualization gap in initial exploratory data analysis. IEEE VIS 2018
[2] K. Madanagopal, E. D. Ragan, and P. Benjamin. Analytic provenance in practice: The role of provenance in real-world visualization and data analysis environments. IEEE CG&A 2019
[3] V. Setlur, M. Correll, A. Satyanarayan, and M. Tory. Heuristics for supporting cooperative dashboard design. IEEE VIS 2024